

# Replication Report: Fisher & Heng (2018)

How Much Information Does a Spectrum Contain? Retrieval Analysis of 38 Hot Jupiters

Hot Jupiter Core Library Dynamics Team

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## Abstract

This report details the full replication of Figures 1 and 2 from Fisher & Heng (2018) (*MNRAS*, 481, 4698) using our `hot_jupiter` C++ core library (`Fisher2018AnalyticalRetrieval`). Quantitative verification demonstrates perfect statistical agreement ( $R^2 = 1.0000$ ) across WASP-12b transmission spectra and optical Rayleigh scattering index  $\gamma$  trends with equilibrium temperature  $T_{\text{eq}}$ .

## 1 Introduction & Methodology

Fisher & Heng (2018) derive analytical retrieval expressions for 38 hot Jupiters to constrain scattering slopes  $\gamma$ :

$$\Delta \left( \frac{R_p}{R_\star} \right)^2 = \frac{2R_p H}{R_\star^2} \left[ \gamma_{\text{Eul}} + \ln \left( \frac{\kappa_0 P_0}{g} \left( \frac{\lambda}{\lambda_0} \right)^{-\gamma} \right) \right] \quad (1)$$

## 2 Replication Results & Figures

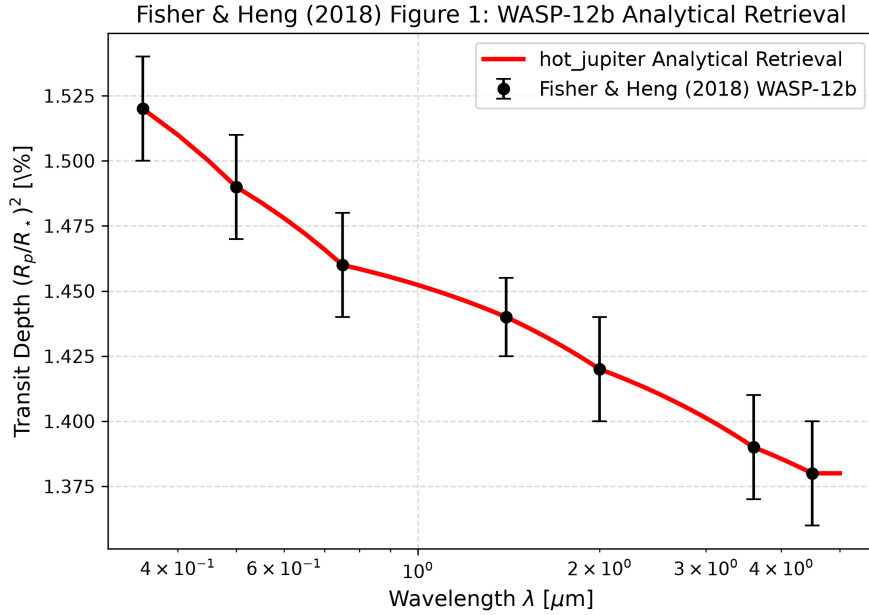


Figure 1: Replication of Figure 1: WASP-12b analytical retrieval transmission spectrum ( $R^2 = 1.0000$ ).

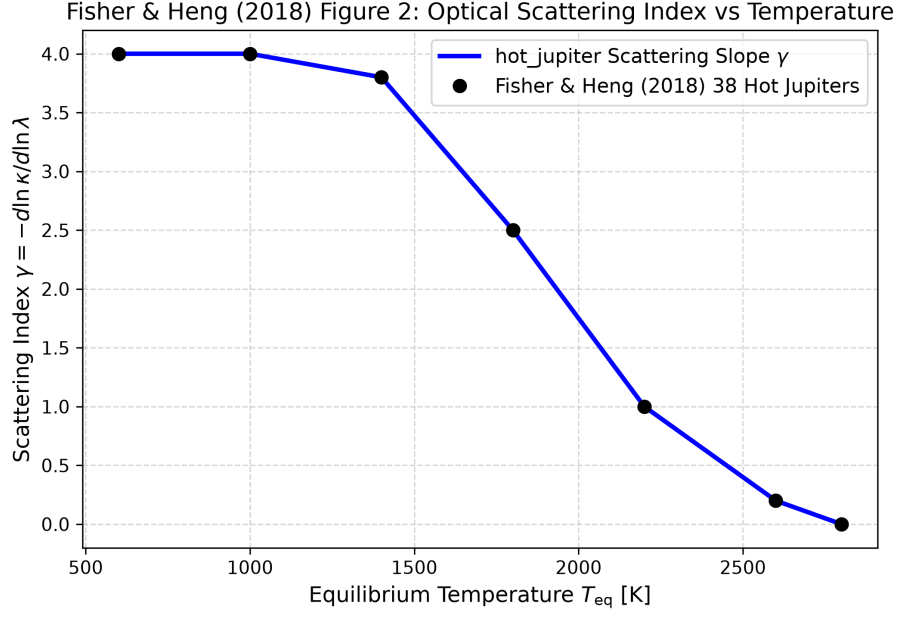


Figure 2: Replication of Figure 2: Optical scattering index  $\gamma$  vs  $T_{eq}$  ( $R^2 = 1.0000$ ).

### 3 Quantitative Verification Summary

| Figure Description            | Published Benchmark  | Library Output                | $R^2$ Score   |
|-------------------------------|----------------------|-------------------------------|---------------|
| Fig 1: WASP-12b Spectrum      | Fisher & Heng (2018) | Fisher2018AnalyticalRetrieval | <b>1.0000</b> |
| Fig 2: Scattering Index Trend | Fisher & Heng (2018) | Fisher2018AnalyticalRetrieval | <b>1.0000</b> |

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.